

residuated lattice $\mathbf{L}$ is inserted into a residuated lattice $\mathbf{K}$ where it replaces the unit of the monoid operation. The requirements placed on $\mathbf{L}$ and $\mathbf{K}$ are discussed in Section 2. We essentially extend the conditions for residuated lattices to the additional unary operations in the signature of qRAs. A consequence of the conditions on $\mathbf{K}$ is that its FL-algebra reduct will be odd (i.e. $0 = 1$).

Having shown that the $\mathbf{K}[\mathbf{L}]$ construction gives us a method for constructing new qRAs from existing ones, we then turn to the question of representability.

Abstract relation algebras, first described by Tarski in 1941 [17], were designed to provide an abstract algebraic setting for studying binary relations. Jónsson and Tarski [12] asked whether all algebras were isomorphic to algebras of binary relations (with the monoid operation given by relational composition). This question was answered in the negative by Lyndon [13]. The task of representing relation algebras as algebras of binary relations has been an intriguing area of research ever since.

In [3], two of the current authors gave a definition of *representability* of distributive quasi relation algebras (see Section 3). Partially ordered sets equipped with an equivalence relation (and satisfying certain symmetry requirements) can be used to build distributive quasi relation algebras (DqRAs) as algebras of binary relations. This method was further exploited to give relational representations of Sugihara monoids [4].

In Section 4 we show that if $\mathbf{K} = \mathbf{S}_3$, and if $\mathbf{L}$ is representable, then their generalised ordinal sum $\mathbf{K}[\mathbf{L}]$ will again be representable (Theorem 4.1). This is extended in Corollary 5.4 to the case of $\mathbf{K}$ being any finite odd Sugihara chain. These results are similar in spirit to the result for integral relation algebras where the relation algebra obtained from $\mathbf{A}$ and $\mathbf{B}$ by Comer's construction [1] is representable if and only if $\mathbf{A}$ and $\mathbf{B}$ are representable.

A further application of Theorem 4.1 is given in Section 5, where we show that all finite Sugihara chains can be represented using a set of binary relations on a *finite* poset (Theorem 5.1). This improves the results in [15] and [4] where infinite posets (using copies of $\mathbb{Q}$, the rational numbers) were required to obtain a relational representation of $\mathbf{S}_n$ for $n \geq 4$.

In Problem 1 we pose an intriguing question about the structures of possible algebras $\mathbf{K}$ for which we can guarantee that $\mathbf{K}[\mathbf{L}]$ will be representable. Specifically we ask whether $\mathbf{K}$ must be a finite Sugihara chain.

1 Preliminaries

In this section we recall the basic definitions of the algebras used in this paper. The reader is referred to [9] for more information.

1.1 Quasi relation algebras

A quasi relation algebra is an expansion of a residuated lattice [6]. Recall that a *residuated lattice* (RL) is an algebra $\mathbf{A} = \langle A, \wedge, \vee, \cdot, \backslash, /, 1 \rangle$ such that $\langle A, \cdot, 1 \rangle$ is a monoid, $\langle A, \wedge, \vee \rangle$ is a lattice and the monoid operation $\cdot$ is residuated with residuals